

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 — SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – Scenario Based
Weightage:	30% of CIA		

CO Statement: Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)

Student Name:	MADHAVAN S	Reg. No.:	192425113
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ANNEXURE A — SCENARIO BASED (Worked Solutions)

Question 1 — Search Engine Indexing

Scenario: A search engine must group "analyzing", "analysis", and "analytical" under a unified representation to improve retrieval accuracy.

Step 1: Root Identification

The common underlying lexical root/lemma is "analyze" (verb: to examine methodically). Note that "analysis" and "analytical" are historically Greek-derived and do not decompose by simple English suffix-stripping, so a hybrid rule + lexicon approach is required.

Step 2: Morphological Decomposition

Word	Root	Affix	Form Type	POS
analyzing	analyze	-ing	Inflectional (progressive aspect)	Verb
analysis	analy-	-sis (nominalizer)	Derivational (irregular, verb → noun)	Noun
analytical	analy-	-tical (via analytic + -al)	Derivational (noun/verb → adjective)	Adjective

Step 3: Inflectional vs Derivational Differentiation

- analyzing is purely inflectional — it stays a verb and only marks aspect (ongoing action); no change in lexical category or core meaning.
- analysis is derivational — the -sis nominalizer changes the word from verb-like root to noun, denoting the result/process of analyzing. This is an irregular derivational pattern (Greek-origin suffix), not standard English suffixation.
- analytical is derivational — built from "analysis"/"analytic" + -al, converting the concept into an adjective describing a method or approach.

Step 4: Normalization for Indexing

- For retrieval purposes, all three surface forms are mapped to a single normalized index key: "analyze" (or lemma "analyze"), so a query on any variant matches documents containing any of the three forms.

Final Output

Word	Morphological Structure	Normalized Root
analyzing	analyze + ing (INFL)	analyze
analysis	analy + sis (DERIV, irregular)	analyze
analytical	analy + tic + al (DERIV)	analyze

Question 2 — Sentiment Analysis Platform

Scenario: A sentiment analysis system must interpret "disagree", "agreement", and "agreeable" consistently despite differing word formation.

Step 1: Base Form Identification

The shared base is "agree" (verb, to have the same opinion / to be suitable).

Step 2: Prefix/Suffix Decomposition

Word	Prefix	Base	Suffix	Resulting POS
disagree	dis-	agree	—	Verb (meaning reversed)
agreement	—	agree	-ment	Noun
agreeable	—	agree	-able	Adjective

Step 3: Classification of Each Transformation

- disagree: derivational — prefix dis- attaches to a verb and returns a verb, but reverses polarity (agree → not agree). This is a semantically critical negation for sentiment systems.
- agreement: derivational — suffix -ment converts the verb agree into a noun denoting the state or outcome of agreeing.
- agreeable: derivational — suffix -able converts agree into an adjective meaning "pleasant/willing to agree", shifting from an action to a quality/disposition.

Step 4: Semantic Impact on Sentiment Interpretation

- disagree carries negative/contrastive sentiment polarity relative to the base "agree" — a stemmer that strips "dis-" naively (treating it as noise) would incorrectly merge opposite-polarity reviews.
- agreement is largely neutral-to-positive, describing a state reached through consensus.
- agreeable is typically positive, describing a favorable quality or disposition.

Final Root Extraction and Classification

Word	Root	Affix(es)	Type	Sentiment Implication
disagree	agree	dis- (prefix)	Derivational	Negative / contrastive
agreement	agree	-ment (suffix)	Derivational	Neutral–Positive
agreeable	agree	-able (suffix)	Derivational	Positive

Question 3 — News Aggregation Text Analytics

Scenario: A news aggregation pipeline must standardize "govern", "government", and "governance" for topic modeling and clustering.

Step 1: Root Identification

Common root: "govern" (verb, to rule/administer/control).

Step 2: Decomposition into Root + Affix

Word	Root	Suffix	Form Type	POS
govern	govern	—	Base form	Verb
government	govern	-ment	Derivational (verb → noun; institution/entity)	Noun
governance	govern	-ance	Derivational (verb → noun; process/manner)	Noun

Step 3: Derivational Layer Analysis

- government (-ment) typically denotes the institution/body that governs, or the system/act of governing as a concrete entity.
- governance (-ance) typically denotes the process, method, or manner of governing/administration — a subtler, more abstract nuance than government, important to preserve during clustering since news topics about "governance reform" differ from "government elections".
- Both are derivational nominalizations of the same verb root but are NOT full synonyms — a topic model that merges them too aggressively risks losing this distinction, so normalization should map both to root "govern" for retrieval while retaining the original surface form as a secondary feature for fine-grained topic separation.

Step 4: Normalized Representation

Word	Morphological Breakdown	Normalized Base
govern	govern (root)	govern
government	govern + ment (DERIV)	govern
governance	govern + ance (DERIV)	govern

Question 4 — Document Classification System

Scenario: A document classification system processes "activate", "activation", and "reactivation" to improve semantic grouping and indexing.

Step 1: Base Form Identification

Base verb: "activate" (itself derived historically from "active" (adjective) + -ate verbalizer, but treated as the base for this word family since it is the shortest common form given).

Step 2: Layered Morphological Decomposition

Word	Prefix	Base	Suffix	Layers	POS
activate	—	activate	—	1 layer (base)	Verb
activation	—	activate	-ion	1 derivational layer	Noun
reactivation	re-	activate	-ion	2 layers: prefix + suffix	Noun

Step 3: Effect of Prefix Addition and Suffix Transformation

- activation: suffix -ion converts the verb activate into a noun denoting the act/process of activating — a straightforward single-layer derivational change (verb → noun).
- reactivation: two layers combine — first the prefix re- attaches to activate to mean "activate again" (still conceptually verbal), then -ion nominalizes the whole unit. This is a compositional/layered derivation: [re + [activate + ion]], i.e., the prefix scopes over the entire derived noun "activation", not just the bare root.
- Word-class trajectory: activate (V) → activation (N, single suffix layer) → reactivation (N, prefix + suffix layers) — each step adds a distinct semantic component (process-hood, then repetition) without changing the final noun category from activation to reactivation.

Step 4: Unified Base Representation

Word	Morphological Structure	Normalized Root
activate	activate (base verb)	activate
activation	activate + ion	activate
reactivation	re + activate + ion	activate

Question 5 — Search Optimization Module

Scenario: A search optimization module processes "create", "creates", and "creating" to ensure consistent retrieval across grammatical variations.

Step 1: Root Identification

Common root: "create" (verb, to bring into existence).

Step 2: Suffix Pattern Identification

Word	Root	Suffix	Grammatical Feature	Orthographic Change
create	create	—	Base form (infinitive/present)	None
creates	create	-s	3rd person singular present tense	None
creating	creat	-ing	Present participle / progressive aspect	Final -e dropped before -ing

Step 3: Classification of Each Transformation

- creates: inflectional — adds -s to mark 3rd person singular present tense; the verb remains a verb, no new lemma is formed.
- creating: inflectional — adds -ing to mark progressive/continuous aspect, with a standard orthographic rule (drop final silent -e before a vowel-initial suffix); still the same verb, same lemma.
- Both transformations are purely grammatical/tense-aspect marking with no change in part of speech or core meaning — classic inflectional morphology, unlike Q1–Q4 which involved derivational category shifts.

Step 4: Normalized Output for Retrieval

Word	Morphological Breakdown	Normalized Root
create	create (base)	create
creates	create + s (INFL, 3sg present)	create

Word	Morphological Breakdown	Normalized Root
creating	creat + ing (INFL, progressive, e-deletion)	create

All three forms normalize to the single index key "create", ensuring a query for any tense/aspect variant retrieves the full set of matching documents without semantic loss (since inflection does not alter meaning, only grammatical marking).

End of Worked Solutions